

FINTRUST DIGITAL BANK

Week 2 Data Quality Assessment

Data Quality Documentation

Track: Data Analytics

Project: FinTrust Financial Intelligence & Digital Banking Support Solution

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Assessment period: Week 2 - Analyse & Prepare

Document purpose

This document summarizes the overall quality of the FinTrust Customer and Transaction datasets before downstream SQL, Python and Power BI analysis. It records the issues identified, their analytical impact, the actions taken, and the key findings from the quality checks.

1. Purpose and Scope

The assessment reviewed the cleaned FinTrust customer and transaction datasets used for Week 2 analysis. The objective was to confirm that the data was sufficiently complete, consistent and structurally reliable for analysis, while documenting issues that could affect interpretation.

- Customer dataset: 1,500 records and 12 fields; Customer_ID is the unique customer identifier.
- Transaction dataset: 12,000 records; Transaction_ID is the unique transaction identifier and Customer_ID links each transaction to a customer.
- Assessment areas: missing values, duplicates, identifiers, data types, inconsistent values, unusual values, potential outliers, table relationships and analysis-impacting issues.

2. Executive Summary

Overall assessment

The FinTrust datasets were generally suitable for analysis after limited preparation. The customer dataset was complete, while the main transaction-data issues were missing Device_Type and Location values and the need to prepare the transaction date/time field for analysis. No duplicate primary-key values, orphan customer keys, or non-positive transaction amounts were identified.

Quality Area	Finding	Potential Impact	Action / Decision
Customer completeness	No missing values identified in the customer dataset.	Low	No corrective action required.
Device_Type	96 missing values identified.	Moderate for device-level analysis	Missing values standardized as N/A in the cleaned dataset.
Location	96 missing values identified.	Moderate for location-level analysis	Missing values standardized as N/A in the cleaned dataset.
Combined missingness	4 rows were missing both Device_Type and Location; 188 rows had at least one of the two fields missing.	Moderate for analyses using these dimensions	Retained rows; missing categories were standardized rather than deleted.
Duplicate identifiers	Customer_ID was unique across 1,500 customers; Transaction_ID was unique across 12,000 transactions.	Low	No duplicate removal required.
Customer_Name	Only 180 distinct names across 1,500 customer records; names are not unique identifiers.	High if names were used for joins/counts	Use Customer_ID for joins and distinct-customer measures.
Customer-transaction relationship	All 12,000 transaction Customer_ID values matched a customer; all 1,500 customers appeared at least once in transactions.	Positive finding	Use Customer_ID as the one-to-many relationship key.
Transaction amounts	No non-positive values; observed range NGN 100.17 to NGN 693,454.45.	Potential high-value outliers may affect averages	Retain values and review distribution during EDA.

Quality Area	Finding	Potential Impact	Action / Decision
Transaction date/time	Original combined datetime required analysis preparation.	Could limit trend/time analysis if left as text	Split into Transaction_Date and Transaction_Time and standardized types.

3. Detailed Data Quality Assessment

3.1 Customer Dataset

- Record count: 1,500 customer rows.
- Customer_ID: 1,500 distinct values, confirming uniqueness at customer level.
- Missing values: none identified in the customer dataset during profiling.
- Customer_Name: not unique; 180 distinct names were observed, so it is descriptive rather than an identifier.
- Age: observed range 18 to 65 years.
- Customer_Segment: four categories - Everyday, Premium, SME and Student.
- Account_Type: Current, Premium and Savings.
- Account_Status: Active, Dormant and Restricted.
- Digital_Engagement_Score: observed range 16.7 to 100.

Assessment conclusion: The customer table was structurally clean and did not require record deletion. The main control decision was to use Customer_ID rather than Customer_Name for joins, counts and customer-level analysis.

3.2 Transaction Dataset

- Record count: 12,000 transaction rows.
- Transaction_ID: 12,000 distinct values, confirming uniqueness at transaction level.
- Customer_ID: 1,500 distinct customer identifiers represented; all transaction keys matched the customer table.
- Device_Type: 96 missing values identified before cleaning.
- Location: 96 missing values identified before cleaning.
- Four transaction rows were missing both Device_Type and Location; 188 rows had at least one of these two fields missing.
- Amount_NGN: no zero or negative transaction amounts were identified. Values ranged from NGN 100.17 to NGN 693,454.45.
- Transaction_Status: four categories - Successful, Failed, Reversed and Pending.
- Risk_Review_Flag: Yes/No synthetic review indicator; it is not a confirmed fraud label.
- Transactions per customer ranged from 1 to 20, with an average of approximately 8 transactions per customer during profiling.

Assessment conclusion: The transaction table was broadly analysis-ready after missing-value standardization and date/time preparation. High transaction values were retained because no evidence showed that they were invalid; they were instead reviewed as part of exploratory analysis.

4. Cleaning and Preparation Actions

Field / Area	Issue	Action Taken	Outcome
Transaction_DateTime	Combined date/time field required separate analytical components.	Split into Transaction_Date and Transaction_Time.	Enabled date-based and time-based analysis in SQL, Python and Power BI.
Device_Type	96 missing values.	Replaced missing entries with the standardized label N/A; no rows deleted.	Preserved transaction records while making missingness explicit.
Location	96 missing values.	Replaced missing entries with the standardized label N/A; no rows deleted.	Preserved transaction records while making missingness explicit.
Relevant customer and transaction fields	Imported fields required consistent analytical types.	Standardized identifiers and categories as text, count fields as whole numbers, measures as numeric, and date/time fields to appropriate types.	Improved compatibility across Excel, MySQL, Python and Power BI.
Cleaned dataset outputs	Prepared files required for downstream tools.	Saved cleaned customer and transaction datasets as CSV UTF-8.	Created analysis-ready files for SQL, Python and Power BI.

5. Impact on Analysis

- Customer-level joins and distinct-customer measures must use Customer_ID, not Customer_Name.
- Device_Type and Location analyses should recognize N/A as a missing/unknown category rather than a substantive customer attribute.
- Transaction amounts are strongly right-skewed in exploratory analysis; high-value transactions can influence averages, so medians/distributions should be considered when interpreting typical values.
- The one-to-many Customer_ID relationship supports integrated customer-behaviour analysis without losing transaction records.
- Prepared Transaction_Date supports January-March 2026 trend analysis in SQL and Power BI.

6. Key Quality-Check Findings

Quality Check	Result	Status
Row counts	Customer = 1,500; Transaction = 12,000	Passed
Customer key uniqueness	1,500 unique Customer_ID values	Passed
Transaction key uniqueness	12,000 unique Transaction_ID values	Passed
Orphan transaction keys	None detected	Passed

Quality Check	Result	Status
Customer missing values	None detected	Passed
Device_Type missing values	96 before cleaning	Issue documented and handled
Location missing values	96 before cleaning	Issue documented and handled
Non-positive Amount_NGN	None detected	Passed
Relationship coverage	All 1,500 customers appeared at least once in transactions	Passed
Date/time readiness	Date and time separated and standardized	Passed after preparation

7. Assumptions and Limitations

- All FinTrust customer, transaction and risk-review information is synthetic and created for educational purposes.
- Risk_Review_Flag must not be interpreted as confirmed fraud or used as a real banking risk decision.
- The N/A label preserves records but does not recover the original Device_Type or Location information, so detailed analysis of those fields remains partially limited.
- Some tools may automatically interpret the literal text N/A as a null value when importing CSV files; import settings should be reviewed if the distinction matters.
- Potential transaction-amount outliers were retained because the quality assessment found no evidence that they were data-entry errors.

8. Conclusion

The FinTrust Customer and Transaction datasets were assessed as suitable for Week 2 Data Analytics work after targeted preparation. The customer data was complete and structurally consistent. The transaction dataset contained limited missingness in Device_Type and Location, which was standardized without deleting records, and the transaction datetime field was prepared for time-based analysis. Identifier uniqueness and table relationships were validated, and no orphan customer keys or non-positive transaction amounts were detected. These checks provided a reliable foundation for the subsequent SQL analysis, Python exploratory analysis and Power BI dashboard development.

Overall quality status

Suitable for analysis after documented cleaning and preparation. Remaining limitations are transparent and manageable within the educational scope of the FinTrust project.